The documentation and process conversion measures necessary to comply with this document shall be completed by 12 June 2019.

INCH-POUND

MIL-PRF-19500/642E
w/AMENDMENT 1
12 March 2019
SUPERSEDING
MIL-PRF-19500/642E
6 June 2018

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
DUAL, COMMON CATHODE OR ANODE CENTER TAP, ULTRAFAST,
CASE MOUNT THROUGH HOLE PACKAGE, STANDARD AND REVERSE POLARITY,
TYPES 1N6762 THROUGH 1N6765, QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, dual ultrafast, power rectifier diodes in a center-tap configuration. Four levels of product assurance (JAN, JANTX, JANTXV and JANS) are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outline. The device package for the encapsulated device types is a TO-254AA in accordance with figure 1.

1.3 Maximum ratings.

Types	V_{RWM} (1) I_D = 10 µA dc	I_F (1) (2) T_C =+100°C	I_{FSM} (1) tp= 8.3 ms	$R_{\theta JC}$ (1)	$R_{\theta JA}$ (1)	T_{STG} and T_J
	V dc	A dc	A(pk)	°C/W	°C/W	°C
1N6762, 1N6762R	50	12	165	2.0	40	-65 to +150
1N6763, 1N6763R	100	12	165	2.0	40	
1N6764, 1N6764R	150	12	165	2.0	40	
1N6765, 1N6765R	200	12	165	2.0	40	

(1) Each individual diode.
(2) Derate at 240 mA/°C above T_C = +100°C.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics. Unless otherwise specified, characteristics are each diode at +25°C.

Types	V_{F1} I_F = 6 A dc	V_{F2} I_F =12 A dc	I_{R1} (see 1.3) V_R = 0.8 V_{RWM}	I_{R2} (see 1.3) V_R = 0.8 V_{RWM} T_C = +100°C	trr	C_J V_R = 5 V f = 1 MHz
	V dc	V dc	µA dc	µA dc	ns	pF
1N6762, 1N6762R	0.95	1.05	10	500	35	300
1N6763, 1N6763R	0.95	1.05	10	500	35	300
1N6764, 1N6764R	0.95	1.05	10	500	35	300
1N6765, 1N6765R	0.95	1.05	10	500	35	300

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "1N".

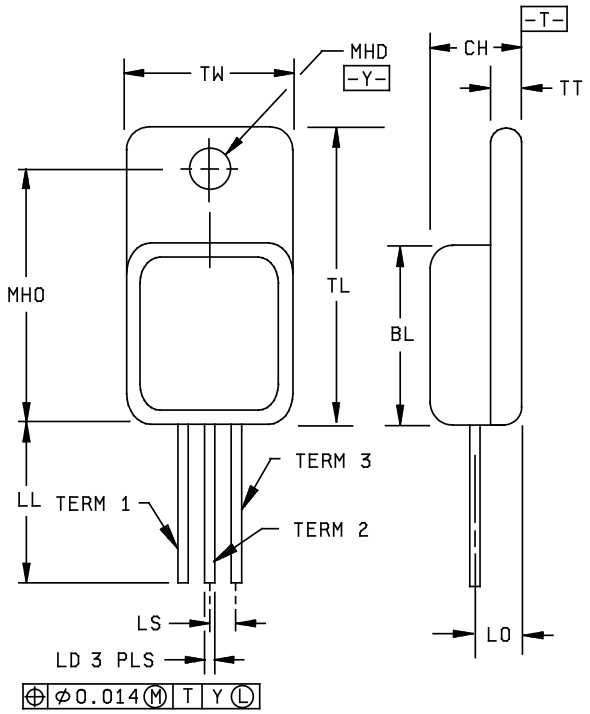
1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are as follows: "6762", "6763", "6764", and "6765".

1.5.3 Suffix symbols. The following suffix symbol(s) are incorporated into the PINs for this specification sheet.

	A blank suffix indicates a standard polarity package configurations as shown on figure 1.
R	An R suffix indicates reverse polarity packaging of the diode.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

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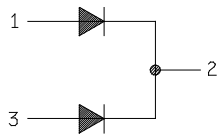


Dimensions				
Symbol	Inches		Millimeters	
	Min	Max	Min	Max
BL	.535	.545	13.59	13.84
CH	.249	.260	6.32	6.60
LD	.035	.045	0.89	1.14
LL	.510	.570	12.95	14.48
LO	.150 typ		3.81 typ	
LS	.150 bsc		3.81 bsc	
MHD	.139	.149	3.53	3.78
MHO	.665	.685	16.89	17.40
TL	.790	.800	20.07	N
TT	.040	.050	1.02	1.27
TW	.535	.545	13.59	13.84

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. All terminals are isolated from case.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

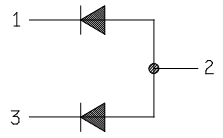
SCHEMATIC

1N6762, 1N6763, 1N6764, 1N6765



Terminal	Description
1	Anode 1
2	Cathode
3	Anode 2

1N6762R, 1N6763R, 1N6764R, 1N6765R



Terminal	Description
1	Cathode 1
2	Anode
3	Cathode 2

FIGURE 1. Physical dimensions and configuration (TO-254AA, isolated).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QPDSIS-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Polarity. Polarity and terminal configuration shall be in accordance with figure 1 herein.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

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3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E inspection. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (quality levels JANTX, JANTXV, and JANS only). Screening shall be in accordance with appendix E, table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	JANS level	JANTX and JANTXV levels
3c (1)	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9 and 10	Not applicable	Not applicable
11	I_{R1} and V_{F1}	I_{R1} and V_{F1}
12	See 4.3.1, t = 240 hours	See 4.3.1, t = 48 hours
13	Subgroups 2 and 3 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial value or ±2.5 µA, whichever is greater; $\Delta V_{F1} \leq$ ±100 mV.	Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial value or ±2.5 µA whichever is greater; $\Delta V_{F1} \leq$ ±100 mV.

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3a, JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. The power burn-in of devices shall be performed in accordance with condition A of method 1038 of MIL-STD-750. The power burn-in conditions shall be as follows: T_C = +125°C; V_R = 0.8 of rated V_{RWM} (see 1.3).

* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD-750, using the guidelines in that method for determining I_M, I_H, t_H, t_{MD}, t_{SW}, (V_C and V_H where appropriate). See table II, subgroup 4 herein.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I, herein. Delta measurements shall be in accordance with the applicable steps of table III herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VIA (JANS) and table E-VIB (JAN, JANTX and JANTXV) of MIL-PRF-19500 and as follows. Delta measurements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Quality level JANS (table E-VIA of MIL-PRF-19500).

	Subgroup	Method	Condition
*	B4	1037	I_F or I_O = 1.25 A to 10 A; ΔT_J = +85°C minimum.

4.4.2.2 Quality levels JAN, JANTX and JANTXV (table E-VIB of MIL-PRF-19500).

	Subgroup	Method	Condition
*	B3	1037	I_F or I_O = 1.25 A to 10 A; ΔT_J = +85°C minimum. For irradiated devices, include t_{rr} as an end-point measurement.
	B5	4081	$R_{\theta JC}$ (maximum) = 2.0°C/W.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500 and as follows. Delta measurements shall be in accordance with the applicable steps of table III herein.

	Subgroup	Method	Condition
	C2	2036	Test condition A, 5 pounds (2.27 Kg), t = 15 seconds ±3 seconds.
	C5		Not applicable.
*	C6	1037	I_F or I_O = 1.25 A to 10 A; ΔT_J = +85°C minimum. For irradiated devices, include t_{rr} as an end-point measurement.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and table II. Delta measurements shall be in accordance with the applicable steps and footnotes of table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. The conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

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* TABLE I. Group A inspection. 1/ 2/

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.3.2	$Z_{\theta JX}$			°C/W
* Breakdown voltage	4021	I_R = 10 µA dc, pulsed	V_{BR}			V dc
1N6762, 1N6762R				50		
1N6763, 1N6763R				100		
1N6764, 1N6764R				150		
1N6765, 1N6765R				200		
Forward voltage	4011	Test condition B I_F = 6 A dc, pulsed 3/ I_F = 12 A dc, pulsed 3/	V_{F1} V_{F2}		0.95 1.05	V dc V dc
Reverse leakage current	4016	DC method; pulsed V_R = 0.8 of V_{RWM} (see 1.3)	I_{R1}		10	µA dc
Subgroup 3						
High temperature operation:		T_C = +100°C				
Reverse leakage current	4016	DC method; pulsed V_R = 0.8 of V_{RWM} (see 1.3)	I_{R2}		500	µA dc
Low temperature operation:		T_A = -55°C				
Forward voltage	4011	Test condition B I_F = 12 A dc, pulsed 3/	V_{F3}		1.25	V dc
Subgroup 4						
Scope display evaluation	4023	4/				
Reverse recovery time measurements	4031	Condition B1	t_{rr}		35	ns

See footnotes at end of table.

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* TABLE I. Group A inspection – Continued. 1/ 2/

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroups 5 and 6						
Not applicable						
Subgroup 7						
Junction capacitance	4001	V_R = 5 V dc; f = 1.0 MHz	C_J		300	pF

1/ For sampling plan, see MIL-PRF-19500.
2/ Each individual diode.
3/ Pulse test: Pulse width = 300 µs, duty cycle ≤ 2 percent.
4/ The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 50 to 100 µA/division and 50 to 100 V/division. Reverse current over the knee shall be at least 500 µA. Each device may exhibit a slightly rounded characteristic and any discontinuity or dynamic instability of the trace shall be cause for rejection.

TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	500 cycles	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table III herein, steps 1 and 2.	
Subgroup 2			45 devices c = 0
Steady-state reverse bias	1038	Test condition A; t = 1,000 hours, T_C = +125°C V_R = 0.8 of rated V_{RWM} (see 1.3)	
Electrical measurements		See table III herein, steps 1 and 2. For irradiated devices, include t_{rr} as an end-point measurement.	
Subgroup 4			
Thermal impedance curves		See MIL-PRF-19500.	Sample size N/A
Subgroup 5			
Not applicable			

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* TABLE III. Groups B, C, and E delta measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
* 1	Forward voltage	4011	Condition B, I_F = 12 A dc pulsed	ΔV_{F2}		±100 mV dc from initial reading	
2	Reverse leakage current	4016	DC method, pulsed V_R = 80 percent rated V_R (see 1.3)	ΔI_{R1}		100-percent of initial value or ±2.5 µA dc which-ever is greater.	

1/ The electrical measurements for table E-VIA (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 4, see table III herein, steps 1 and 2.
   b. Subgroup 5, see table III herein, steps 1 and 2.
2/ The electrical measurements for table E-VIB (JAN, JANTX and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table III herein, steps 1 and 2.
   b. Subgroup 3, see table III herein, steps 1 and 2.
3/ The electrical measurements for table E-VII of MIL-PRF-19500 are as follows:
   a. Subgroup 6, see table III herein, steps 1 and 2 for all levels.
4/ The electrical measurements for group C table E-IX of MIL-PRF-19500 are as follows:
   a. Subgroup 1, see table III herein, steps 1 and 2.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

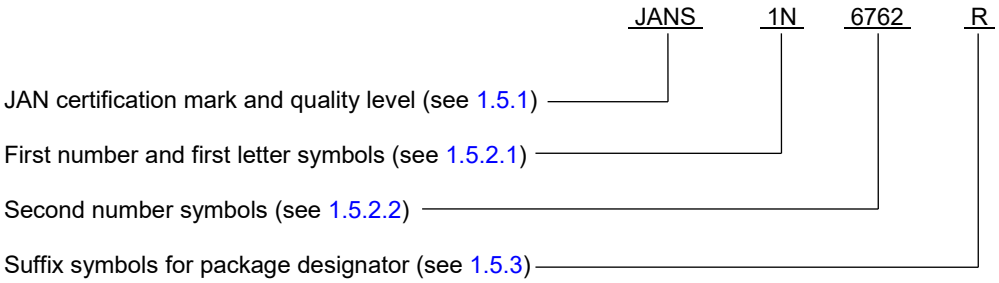
c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QPDSIS-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



6.5 List of PINs. The following is a list of possible PINs available for encapsulated devices covered by this specification sheet.

JAN quality level	JANTX quality level	JANTXV quality level	JANS quality level
JAN1N6762	JANTX1N6762	JANTXV1N6762	JANS1N6762
JAN1N6762R	JANTX1N6762R	JANTXV1N6762R	JANS1N6762R
JAN1N6763	JANTX1N6763	JANTXV1N6763	JANS1N6763
JAN1N6763R	JANTX1N6763R	JANTXV1N676R3	JANS1N6763R
JAN1N6764	JANTX1N6764	JANTXV1N6764	JANS1N6764
JAN1N6764R	JANTX1N6764R	JANTXV1N6764R	JANS1N6764R
JAN1N6765	JANTX1N6765	JANTXV1N6765	JANS1N6765
JAN1N6765R	JANTX1N6765R	JANTXV1N6765R	JANS1N6765R

6.6 Cross reference list. Parts covered by this specification may be used to replace the following commercial PINs:

Preferred types	Commercial types
JANTX1N6762, JANTX1N6762R JANTX1N6763, JANTX1N6763R JANTX1N6764, JANTX1N6764R JANTX1N6765, JANTX1N6765R	1N6762, 1N6762R 1N6763, 1N6763R 1N6764, 1N6764R 1N6765, 1N6765R
JANTXV1N6762, JANTXV1N6762R JANTXV1N6763, JANTXV1N6763R JANTXV1N6764, JANTXV1N6764R JANTXV1N6765, JANTXV1N6765R	1N6762, 1N6762R 1N6763, 1N6763R 1N6764, 1N6764R 1N6765, 1N6765R

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* 6.7 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
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Preparing activity:
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